Exam 2 - MATH-262-01

Instructions: You have 110 minutes to take this exam. Please answer each question in the space provided. Answers without justification may ***earn no credit***. You are permitted your signed prep work journal and a calculator on this exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

1. Consider the ODE

$$x^2y'' - xy' + 2y = 0.$$

(a) Show that $y_1 = x\sin(\ln(x))$ is a solution on $(0, \infty)$.

(b) Use a reduction of order technique to find a second particular solution.

2. Use the method of undetermined coefficients to solve the ODE

$$y'' + 2y' - 24y = 16 - (x+2)e^{4x}.$$

3. Use variation of parameters to solve the IVP

$$2y'' + y' - y = x + 1;\ y(0) = 1,\ y'(0) = 0.$$

4. According to ***Kirchoff's Law***, the impressed voltage on a circuit is the sum of the potential drops across all components. In an LCR circuit, that is

$$V_L + V_R + V_C = V(t).$$

where V_L is the potential drop across the inductor, V_R is the potential drop across the resistor, and V_C is the potential drop across the capacitor. Suppose the circuit has inductance L, resistance R, and capacitance C.

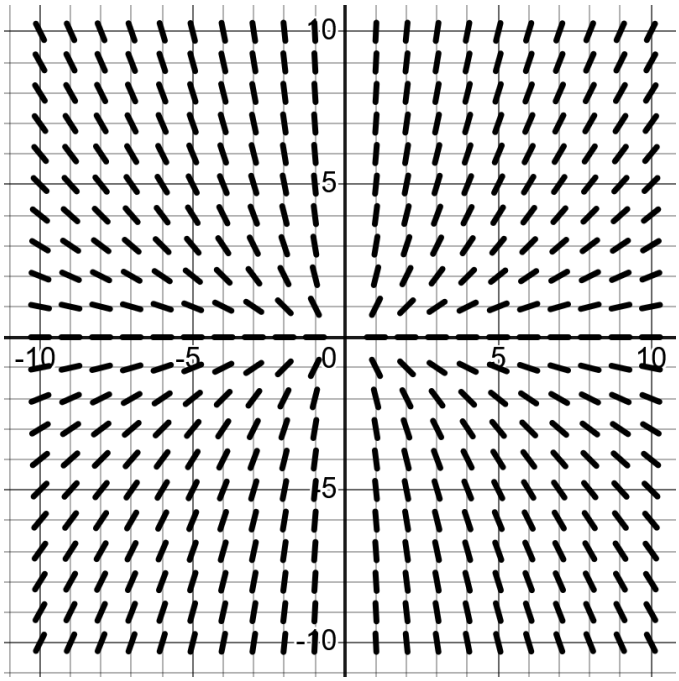
(a) Use the statements below to write a 2nd-order ODE for the charge on the capacitor q at time t.

1. The potential drop across a capacitor is the ratio between its charge and capacitance.
2. Current is the rate of change of the charge on the capacitor.
3. The potnetial drop across a resistor is the product of the resistance and the current.
4. The potential drop across an inductor is the product of the inductance and the rate of change of the current.

(b) Suppose there is no impressed voltage ($E(t) = 0$ for all t). Use the ODE you built in part (b) to construct an algebraic model for the charge on the capacitor at time t.

(c) Suppose the capacitor begins with no charge or current. How does this affect the ODE model you wrote in part (b)? How does this affect the algebraic model you wrote in part (c)?

5. Suppose the slope field below corresponds to a first-order ODE.



(a) Identify all equilibria (nullclines) for the system. Sketch them onto the picture above.

(b) Label at least one nonzero isocline for the system. Sketch it onto the picture above.

(c) Consider the initial condition $y(3) = 1$. Sketch the corresponding trajectory on $(0, \infty)$ onto the picture above.

(d) Classify each of the equilibria you found in part (a) as stable, unstable, or neither on $(-\infty, 0)$. Justify your answer.